

FALCON's Response to RFP Amazon for Supply of Conveyor Automation & Loop CBS System

February 22, 2023



www.falconautotech.com



Kind Attention – Mr. Ashish Sharma

M/s Amazon Saudi Arabia

Offer Ref: F22-00999 ; Date 21.02.2023

Subject – Techno-Commercial Offer for Conveyor Automation + CBS

Dear Team,

We are pleased to submit our Techno-Commercial Offer in response to your RFP

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

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Table of Contents

1. Glossary	6
2. About Falcon	7
3. Handled Parcel Spectrumions Minimum Maximum	14
3.1 Product size Conveyable on the entire system.....	14
3.2 Parcels/Totes to be loaded on Conveyor/Sorter shall have the following characteristics:	15
3.3 Parcel not loadable on the system	15
3.4 Types of Products	16
4. Proposed System Description.....	17
4.1 Objective	17
4.2 Summary of the System	17
a. Main benefits of the proposed solution	18
7. Layout View.....	18
8. Proposed System Capacity	19
a. System Capacity at various areas	19
9. System description	21
a. Inbound lines & Primary diverts	21
b. Trunk line tote conveyor.....	22
c. Parcel Collector lines & SLAM	23
d. Outbound Sorting	24
10. System Components.....	25
a. Powered Roller Conveyor	25
b. Accumulation Conveyor	26
c. PVC Belt Conveyor	27
d. Side Barcode Scanner	29
e. Powered Roller Turns & Merges	30
f. ZTR Curves (Pic for ref.)	30
g. Swivel Wheel Sorting Unit – Falcon	31
h. Spiral Conveyor.....	32
i. VDS Loop (Volume distribution system)	33
j. VDS Chutes.....	34
k. Feedlines.....	35
i. Orientation Conveyor	35

ii. Buffer Conveyor	36
iii. Weighing Conveyor	36
iv. Intelligent Merge	36
v. Main Loop	37
4.3 Cross Belt Sorter -	37
4.4 Servo Roller	38
4.5 Chassis	38
4.6 Carrier Wheels	39
4.7 Non-contact based linear motor drive	39
4.8 Power Transmission	40
4.9 Data Transmission	40
4.10 Carriers positioning System	41
4.11 Technical specifications of Sorter	42
4.12 Scanning & Sensing on Main Sorter Loop	43
1. Automatic Barcode Scanning & Dimensioning System	43
2. Parcel Positioning System.	43
I. Output Chutes	44
i. High-Capacity Chutes	44
ii. Direct Drop Chutes	44
11. Proposed System Technical Details/BOM	45
12. High Availability Server Specifications	46
13. Falcon's WCS CONTROLIT	47
14. Key Component Makes	48
5. Electrical System	48
a. Reference Picture of Power Distribution Panel	49
b. Main Control Panel (Reference)	49
c. Induct Stations Control Panel (Reference)	50
6. Controls Architecture	51
7. IT Architecture (Ref.)	52
8. Principle of Safety	52
15. Automation Management System (AMS)	54
a. Issue Logging	54
b. Preventive Maintenance Check List	55



c.	Predictive Maintenance	55
d.	Training Portal	56
e.	Operational Analysis	56
16.	Program Organisation	57
a.	Program Management	57
17.	Commercial Terms	57
a.	Payment Terms	58
18.	Warranty Period	58
19.	Exclusions	59

1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Parcels Per Hour
3	DBO	Damaged Barcode
4	MEZZ	Mezzanine
5	AC	Alternating Current
6	DC	Direct Current
7	PLC	Programmable Logic Controller
8	IT	Information Technology
9	BOQ	Bill Of Quantity
10	I/O	Input/ Output
11	PDP	Power Distribution Panel
12	PC	Personal Computer
13	UPS	Uninterrupted Power Supply
14	SWEDI	Swivel Wheel Divert

2. About Falcon

THE COMPANY

ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

OUR CORE COMPETENCIES

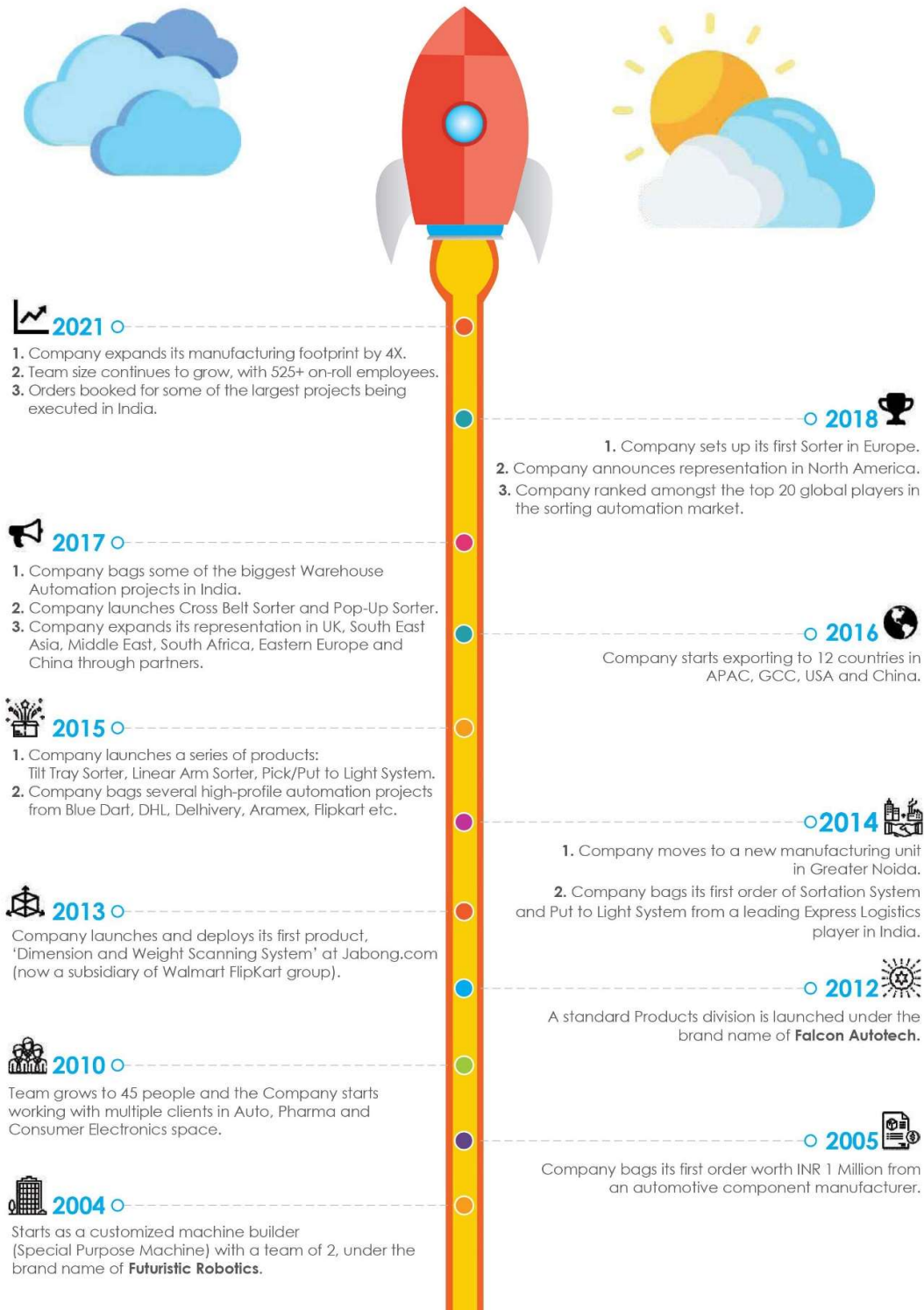
Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.



COMPANY HISTORY



KEY STATISTICS



04

Product Lines



525+

On-Roll Employees



290+

Engineering Team



400000+ Sq Ft

Manufacturing Space



20+

Tier A Clients



15+

Countries with Live Installations



1800+

Total Installed Systems Globally



35

Representations in 4 Continents and 35 Countries



1 Million+

Sorts Per Hour of combined Sorting Capacity
installed Globally

OUR PRODUCT PORTFOLIO

01

SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS



OUR PRODUCT PORTFOLIO

04

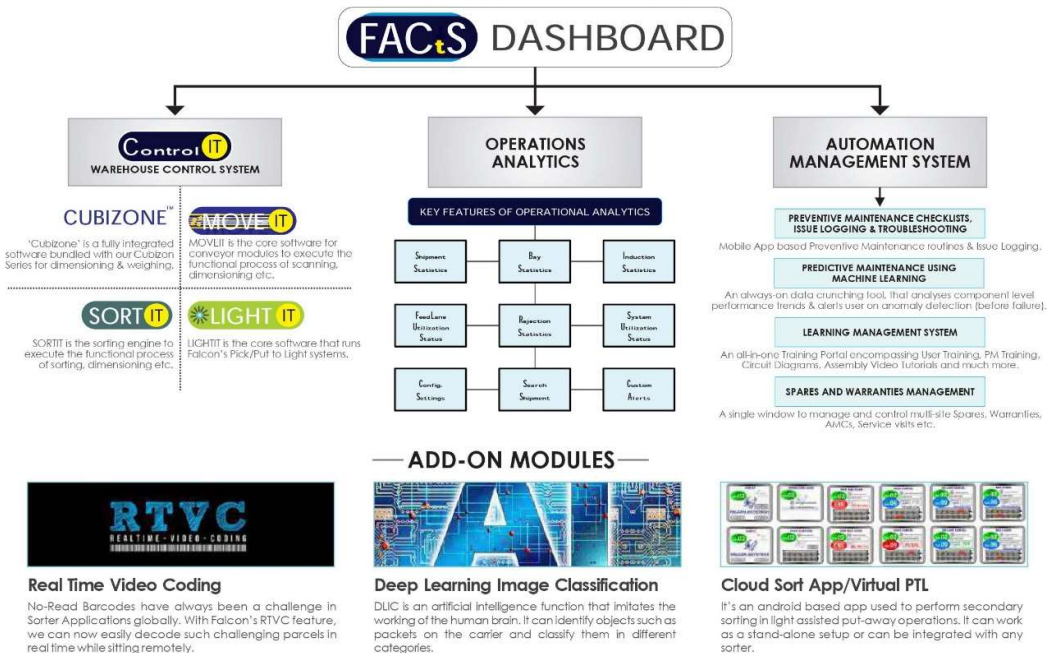
CONVEYOR AUTOMATION SOLUTIONS

BELT CONVEYORS	ROLLER CONVEYORS	SPECIAL APPLICATION SOLUTIONS
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve	◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit	◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor
		

05

SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE



INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



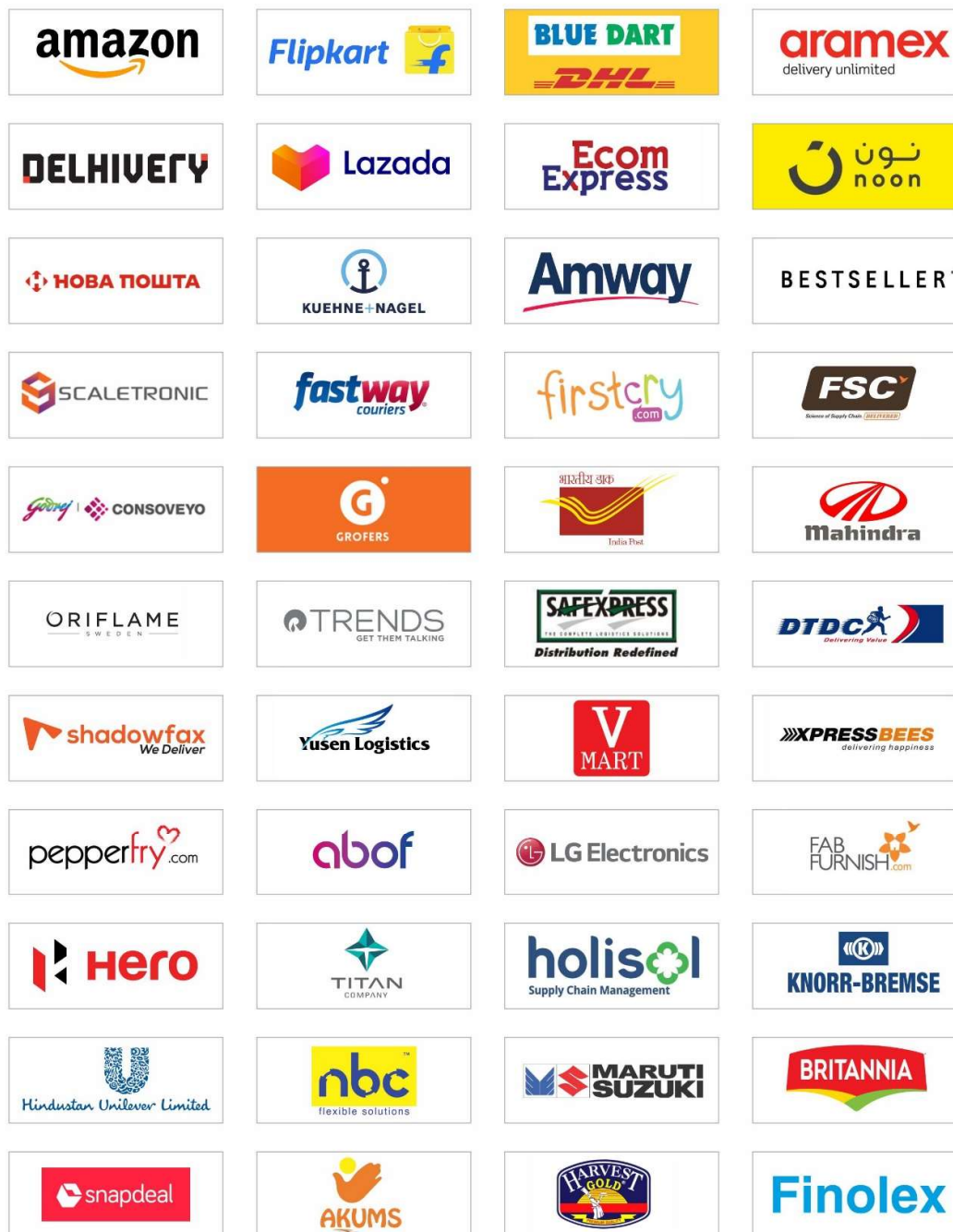
SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



3. Handled Parcel Spectrum

3.1 Product size Conveyable on the entire system.

Inbound Parcel Dimension:

Container	Length (mm)	Width (mm)	Height (mm)	Weight (kg)
Receiving Cartons - Inbound line for cartons				
Minimum	152	150	25	0.1
Maximum	915	635	635	30.0
Empty Cartons - Inbound empty cartons with open flaps				
Minimum	152	150	152	0.1
Maximum	914	737	1,105	3.0
IXD Totes - (Tapered Totes - Outside Dimensions)				
Black Totes	598 ± 2	398.5 ± 1.5	300 ± 1.5	1.81 - 30

Outbound Parcel Dimension:

Container	Length (mm)	Width (mm)	Height (mm)	Weight (kg)
Non-Sort & Multi pack (large) – Collector Non-sort				
Minimum	470	330	203	5.00
Maximum	650	510	470	30.00
Multi pack (small, medium) – Collector Multis (Minimum includes Polybags, Jiffy mailers)				
Minimum	229	152	51	0.45
Maximum	711	470	457	22.68
Single pack – Collector Singles (Minimum includes Polybags, Jiffy mailers)				
Minimum	150	150	3	0.05
Maximum	559	483	305	18.14

3.2 Parcels/Totes to be loaded on Conveyor/Sorter shall have the following characteristics:

1. Centre of Gravity of items must not move during conveyance or sorting.
2. Item must not have magnetic content; otherwise behaviour of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic conveying/sorting system or damaging the system. They must be robust enough to avoid disintegration of container material and loose of contents in the conveying/sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

3.3 Parcel not loadable on the system

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

3.4 Types of Products

Inbound System

Totes/Cartons	Box
	Max: L-915 X W-510 X H-1105 MM
	Min: L-152 X W-150 X H-25 MM

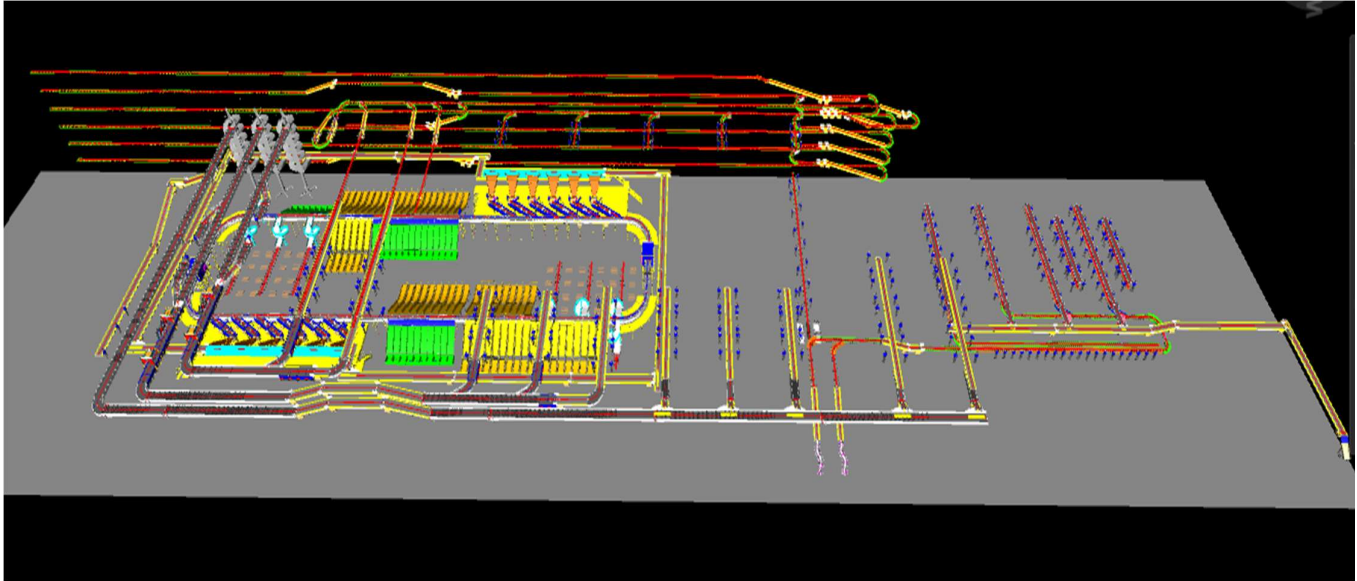
Outbound System

Totes/Cartons	Box
	Max: L-711 X W-510 X H-470 MM
	Min: L-150 X W-150 X H-03 MM

4. Proposed System Description

4.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of the complete automation system to handle Totes & cartons as per specifications shared by Amazon.



4.2 Summary of the System

1. **Infeed (G+0):** At infeed, loose parcels are fed via two flexible conveyors onto set of conveyor leading to primary inbound sorting into 3 destinations.
 - a. **Case receive line**
 - b. **Prep receive line**
 - c. **Each receive line**

Dedicated Trash conveyor is aligned above all four lines to dump waste material in a particular waste zone.

2. **Single (G+4):** Totes transported from G+0,G+1,G+2,G+3 & G+5 are merged into double deck loop highway conveyor at G+4 for further sortation into Singles Zone. Primarily totes get scanned and are diverted in 2 singles lines, diverted totes are then packed into individual parcels on packing table. Shipments are prepared and dropped on packing conveyor running present below tote conveyor, these conveyors merge into highway conveyor and further undergo Auto SLAM(In Amazon scope) process.
3. **Small & Medium Multies (G+4):** Same as single totes, multies are diverted via 5 diverts present at infeed highway conveyor, after packing parcels are fed to packing conveyor which further follows auto SLAM(In Amazon scope) process.
4. **Non-Sort & Large Multies (G+4):** after diverted from highway loop conveyor parcels are packed for further process, Non-sort & Large multi parcel undergoes manual SLAM process and fed onto packing conveyor. These conveyors deposit entire load into three powered spiral conveyor to transport the parcels on G+0 for outbound sorting.

5. **Outbound Sorter:** parcel coming down from 3 spiral conveyors & manually inducted external volume line, get deposited on loop VDS (volume distribution system), Parcels get evenly distributed on Induct line in each induction zone via ARB conveyor. Finally, the parcels are inducted to loop CBS that sorts parcel into final destination (Low volume chutes, medium volume chutes & High volume chute)

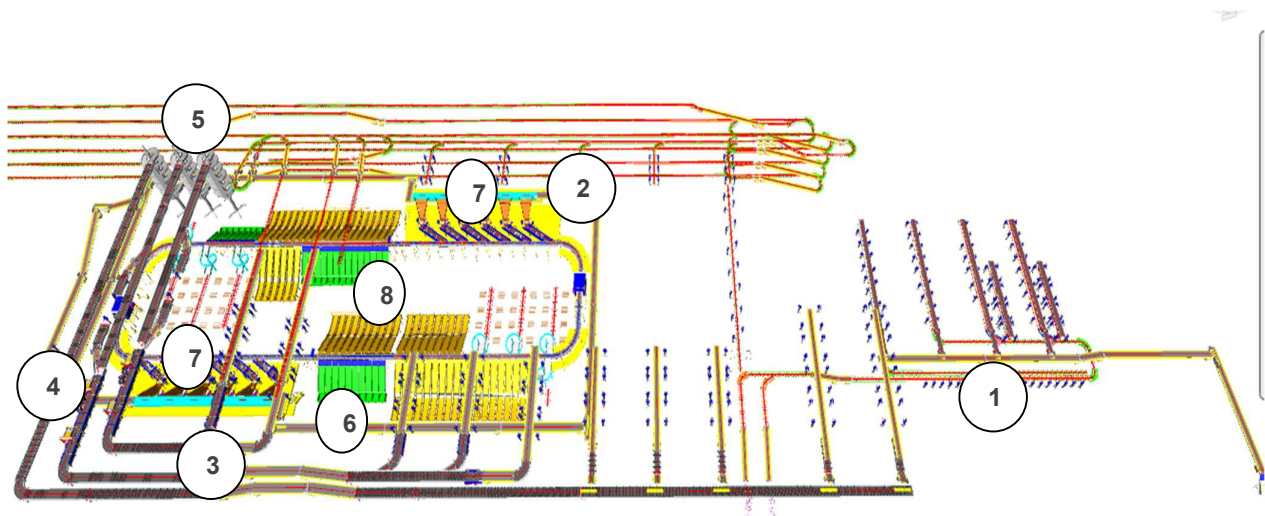
6. **Sort Fail:** SKUs those experience sort fail will get collected in sort fail chute.

a. Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Maximized sorting capacity throughout at dispatch sorter to meet the requirement.
3. Low occupancy of floor space in the building
4. FALCON's system can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear

7. Layout View



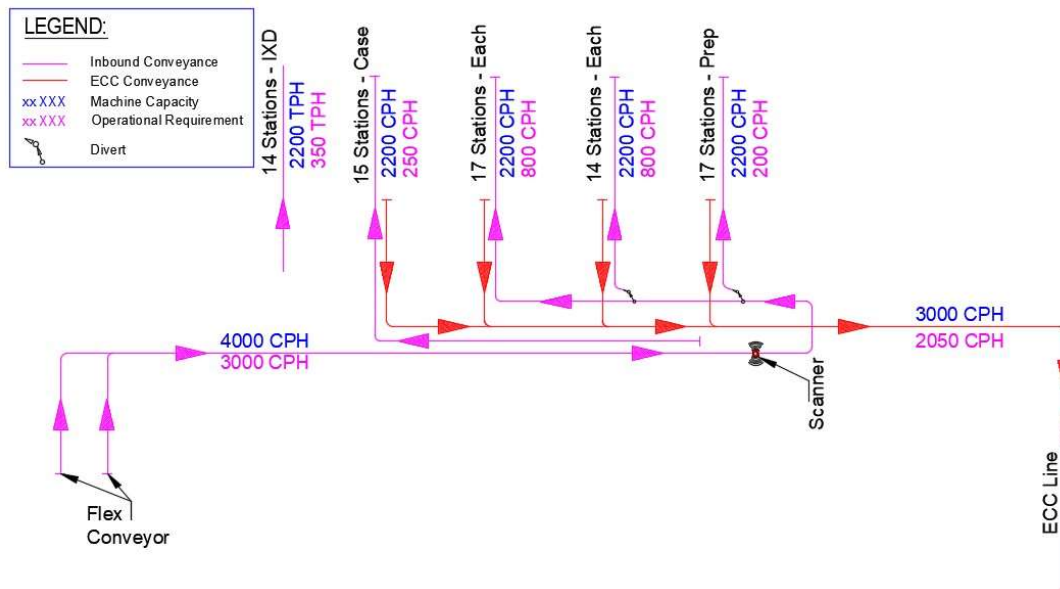
Legend

- 1 – Inbound Zone(G+0)
- 2 – Single, Muties & non sort order consolidation area (G+4)
- 3 – Three Parcel take away conveyor (G+4)
- 4 – Auto SLAM system (G+4) (Amazon Scope)
- 5 – Powered Spiral conveyor (G+4 – G+0)
- 6 – Loop VDS (G+0)
- 7 – Induction zone (G+0)
- 8 – Loop CBS & chutes (G+0)

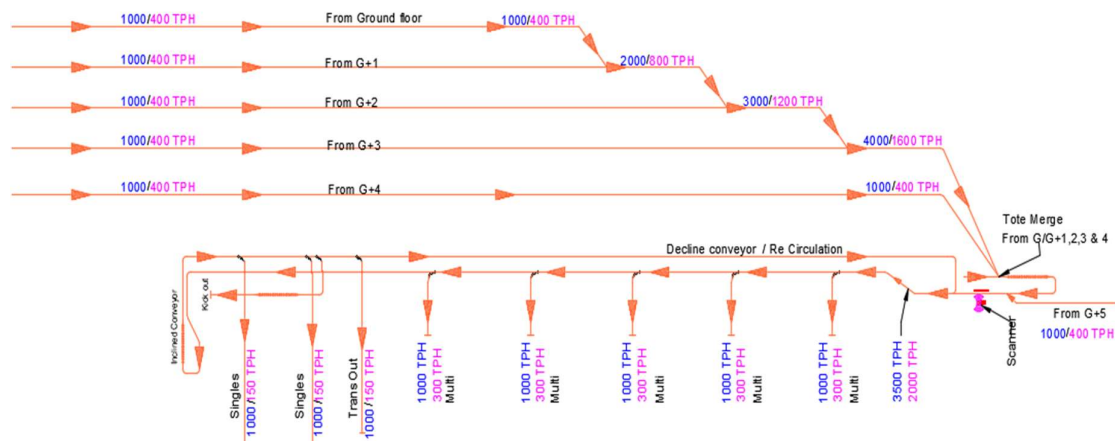
8. Proposed System Capacity

a. System Capacity at various areas

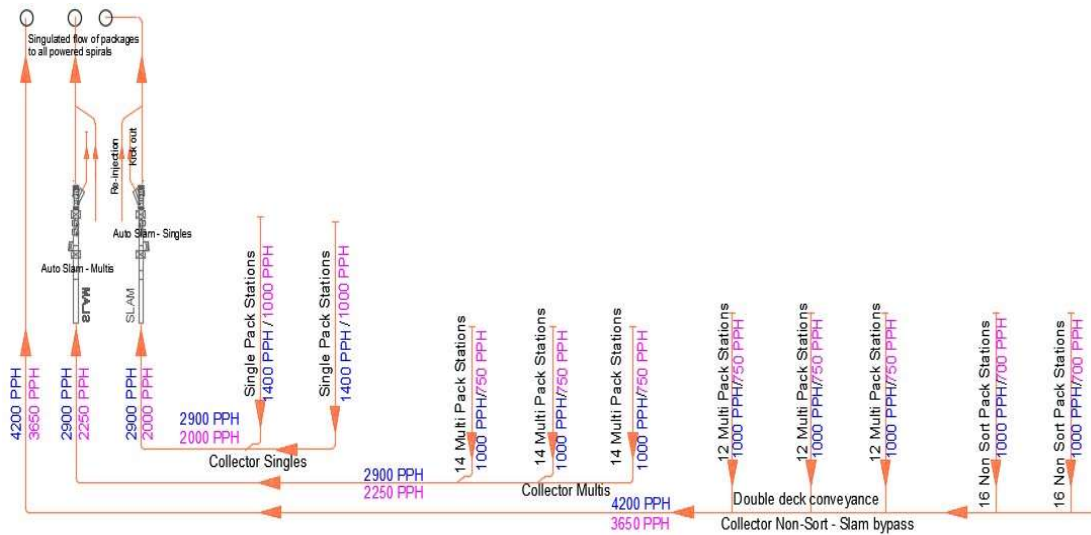
At Inbound



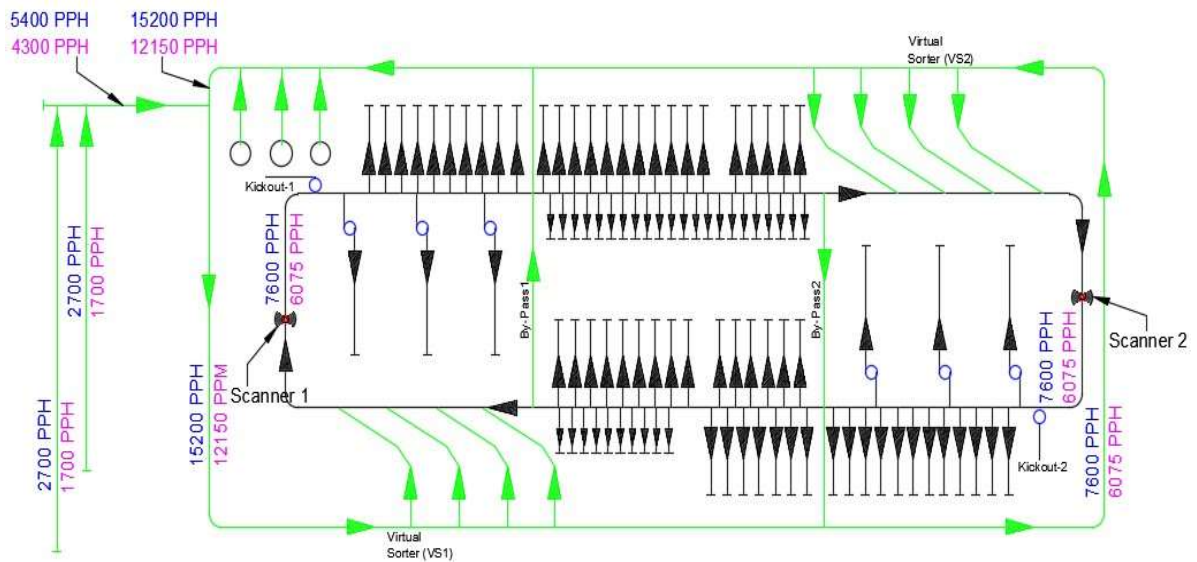
At tote conveyor & sorter MFD



At outbound MFD (G+4)

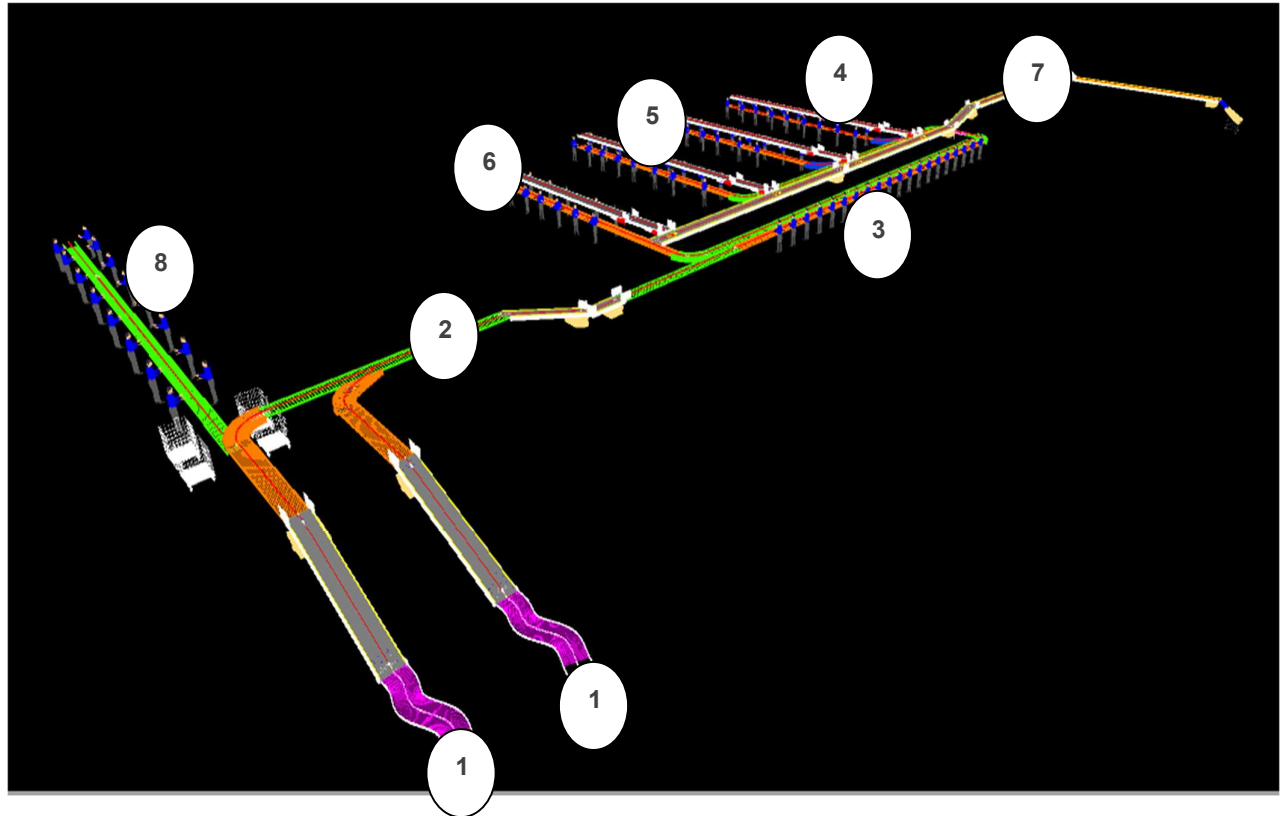


At Outbound CBS (G+0)



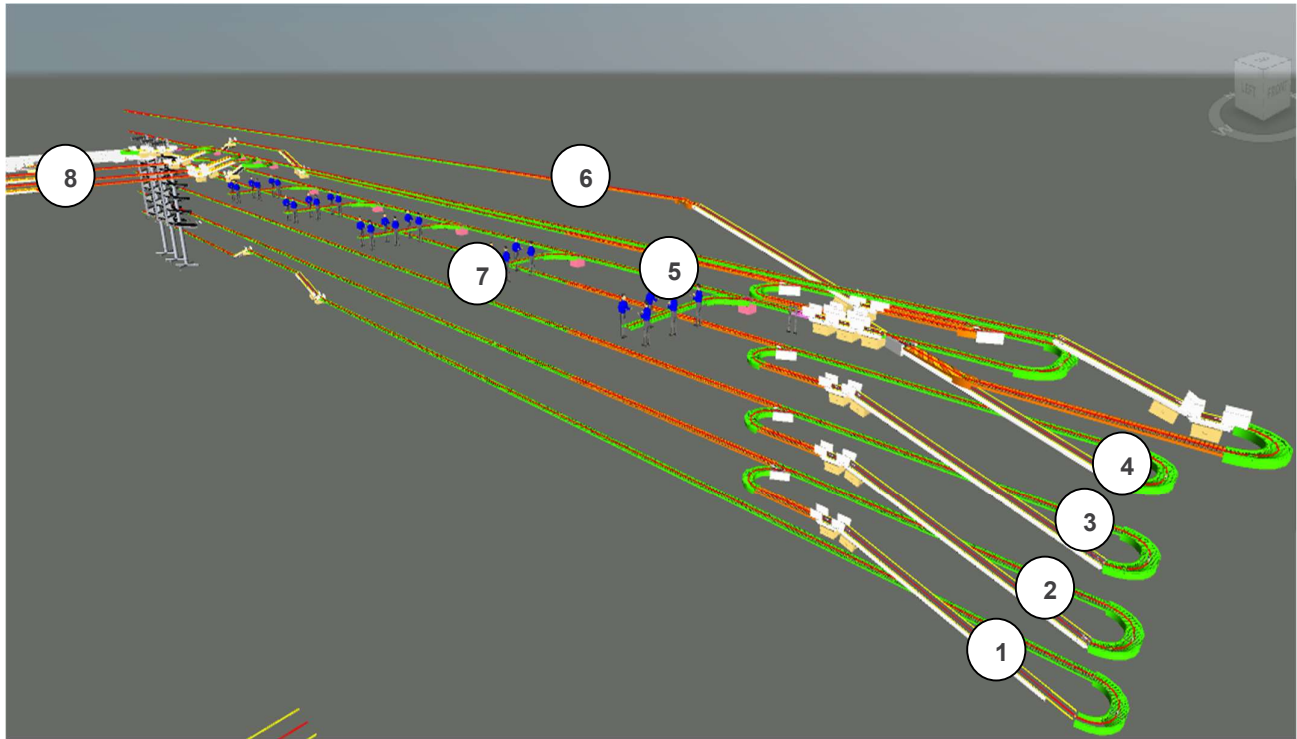
9. System description

a. Inbound lines & Primary diverts



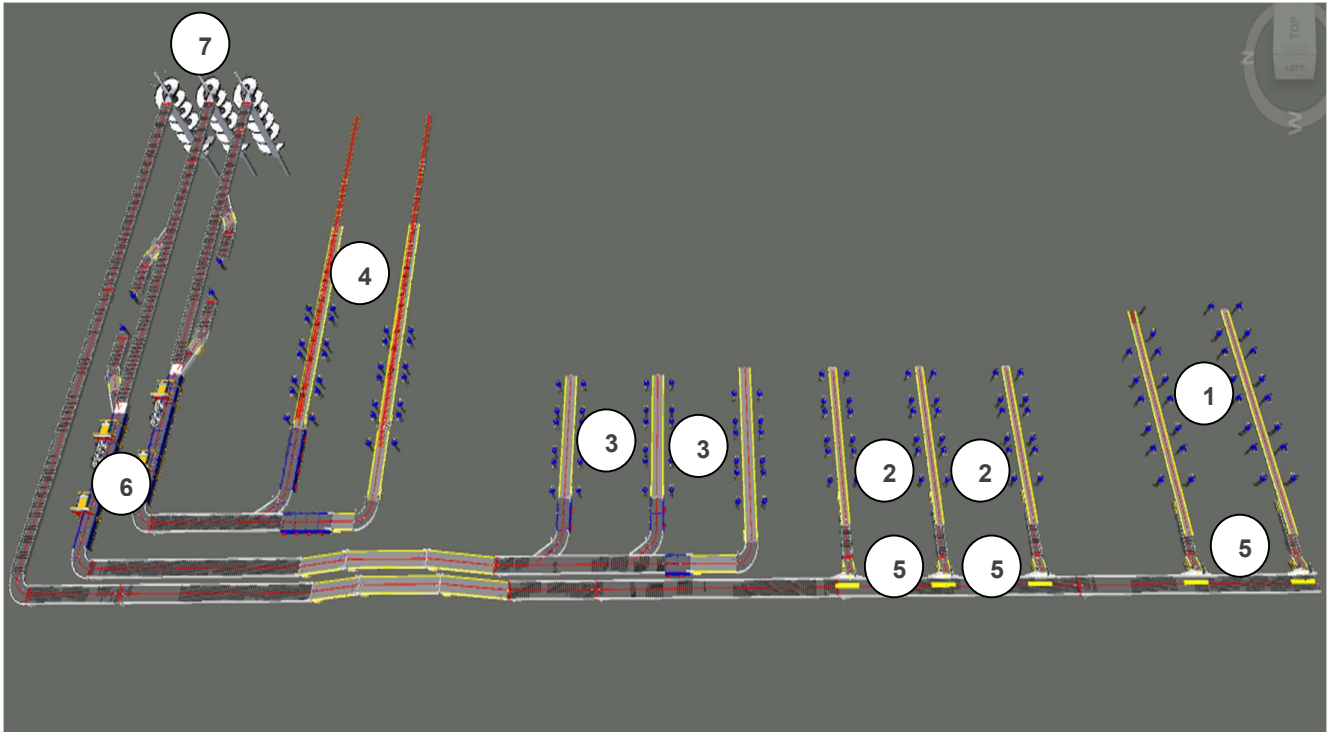
Inbound Legend (G+0)

- 1 – Flexible conveyor
- 2 – Transport receive line
- 3 – Manual scanning zone
- 4 – Prep Line
- 5 – Each Lines
- 6 – Case Line
- 7 – Trash Line
- 8 – IxD conveyor

b. Trunk line tote conveyor**Legend Trunk line G+0,1,2,3,4,5**

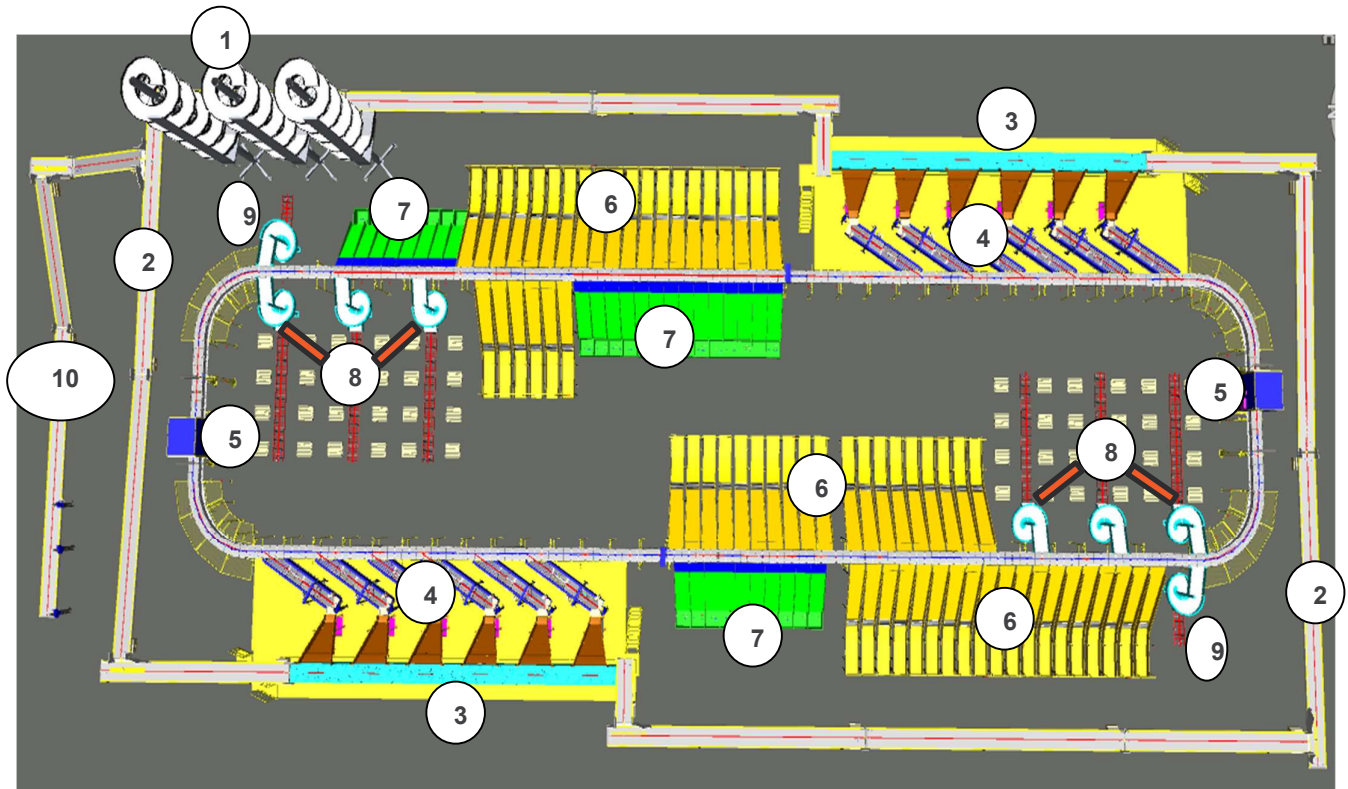
- 1 – Trunk line G+0
- 2 – Trunk line G+1
- 3 – Trunk line G+2
- 4 – Trunk line G+3
- 5 – Trunk line G+4
- 6 – Trunk line G+5
- 7 – Multies divert lines
- 8 – Single divert lines

c. Parcel Collector lines & SLAM

**Legend Collector lines & SLAM**

- 1 – Non sort pack line
- 2 – Multi pack line - large
- 3 – Multi pack line – Small & Medium
- 4 – Single pack line
- 5 – Manual SLAM area
- 6 – Auto SLAM area
- 7 – Spiral conveyor G+4 to G+0

d. Outbound Sorting

**Legend Outbound sorting**

- 1 – Powered spiral conveyor
- 2 – Loop VDS (Volume distribution system)
- 3 – ARB (Activated roller belt)
- 4 – Induction Zone
- 5 – Top Barcode scanner
- 6 – High capacity chutes (yellow)
- 7 – Direct Drop chutes (Green)
- 8 – Gravity spiral chutes
- 9 – Kick out spiral chute
- 10 – External volume line

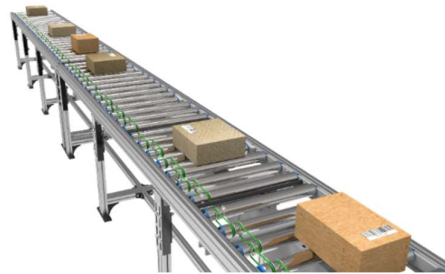
10. System Components

a. Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	800/600
Conveyor Height	mm	Variable as per design shared
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Poly V Belt Drive
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Load capacity/unit length of Conveyor	kg/m	50
Roller Span	mm	Variable
Roller Diameter	mm	50
Pitch of Rollers	mm	124
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

b. Accumulation Conveyor

Falcon's Accumulation conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's accumulation conveyor:

- Low Noise
- Maximum uptime.
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	Mm	600/800
Conveyor Height	Mm	As per Layout requirement
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Round/Poly V Belt Drive
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Load capacity per unit length of Conveyor	kg/m	50
Roller Span	Mm	Approx. 600
Roller Diameter	Mm	Approx. 50
Pitch of Rollers	Mm	75
Zone Length	Mm	680
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Steel guides at 50mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

c. PVC Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	600/800/1000/1200
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Tensioning Rollers	Spec	Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-UCT205D1, MOC-MS, Zinc Plated -25 microns
Take Up Rollers	Spec	Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-6305-LLU, MOC-MS, Zinc Plated -25 microns
Type of Drive	Type	Rubberized

Angle of wrap/Contact - Belt with Drive pulley	Deg	>180 deg
Motor Shaft	MOC	EN9 Shaft material,
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradly
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

d. Side Barcode Scanner

System is integrated with automated inline cameras for multi side scanning with high accuracy.



Specification	UOM	Remark
Manufacturer Name	Name	Cognex/Sick
Barcode Type	Type	1D
Module Size	Mils	12
Number of Codes	Nos	1
Location	Side	Left, Right, Front, back
Orientation	Type	unidirectional
Color of bars	Color	Black
Under Foil	yes/no	No
Code Length	Mm	50
Code Height	Mm	15

e. Powered Roller Turns & Merges



f. ZTR Curves (Pic for ref.)



g. Swivel Wheel Sorting Unit – Falcon

Falcon's Swivel wheel divert is used in system for the smooth transfer of products.

Some Salient Features of Falcon's Swivel Wheel Divert:

- Low Noise
- Maximum uptime.
- Electric based rotation of wheels.
- Minimal maintenance
- High safety standards.

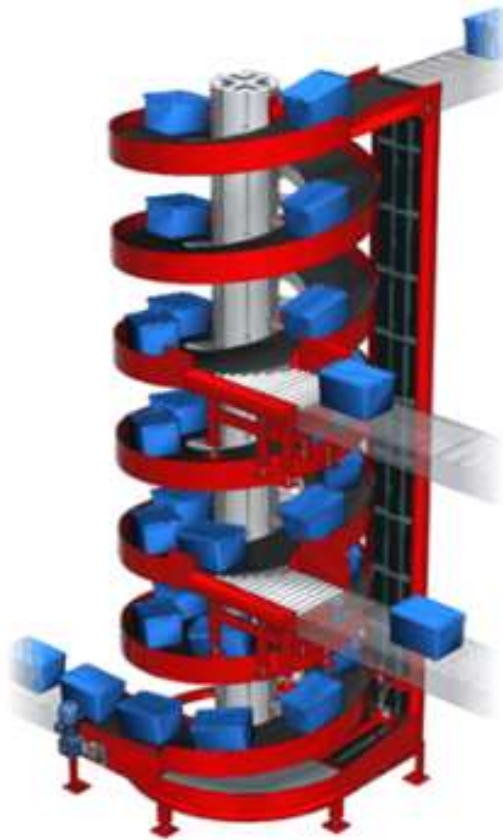


Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Type - Equipment	Type	Merge/Sorting
Type - Operation	Type	Sorting
Mechanism for Swivel	Type	Electric servo motor
Roller drive	kW	Powered smart wheel
No. of divert sections	Yes/No	1 or 2
Max. product weight (kg/m2)	kg/m2	25
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

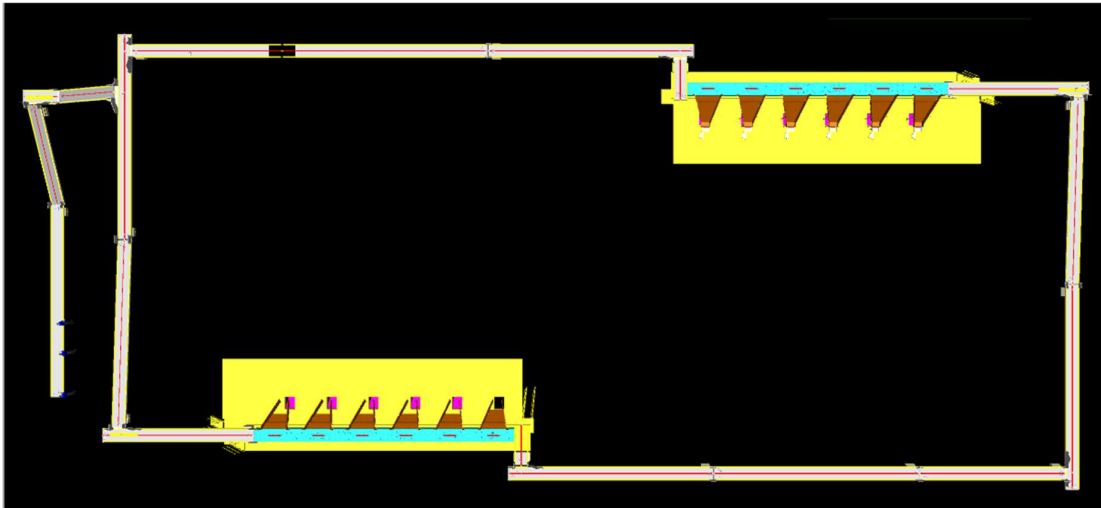
h. Spiral Conveyor

The Spiral conveyor is **a compact and versatile elevating solution**. Spiral can be designed to carry product between floors and machines or over aisles. Designed with a uniform slope, the Spiral conveyor ensures smooth operation without disturbing the product.

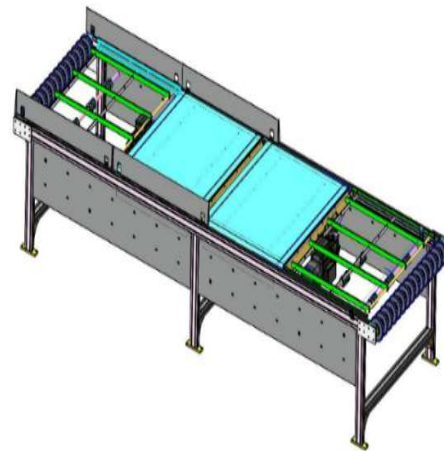
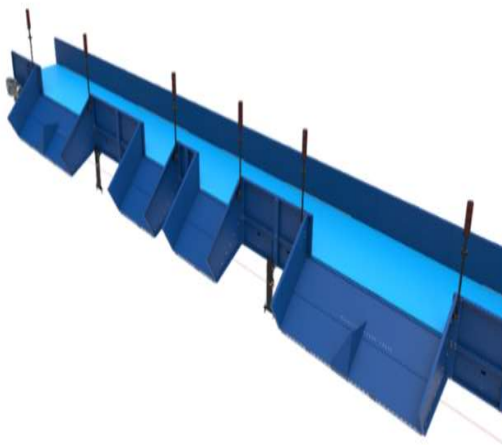
Belt Make – Apollo/Ambaflex



i. VDS Loop (Volume distribution system)



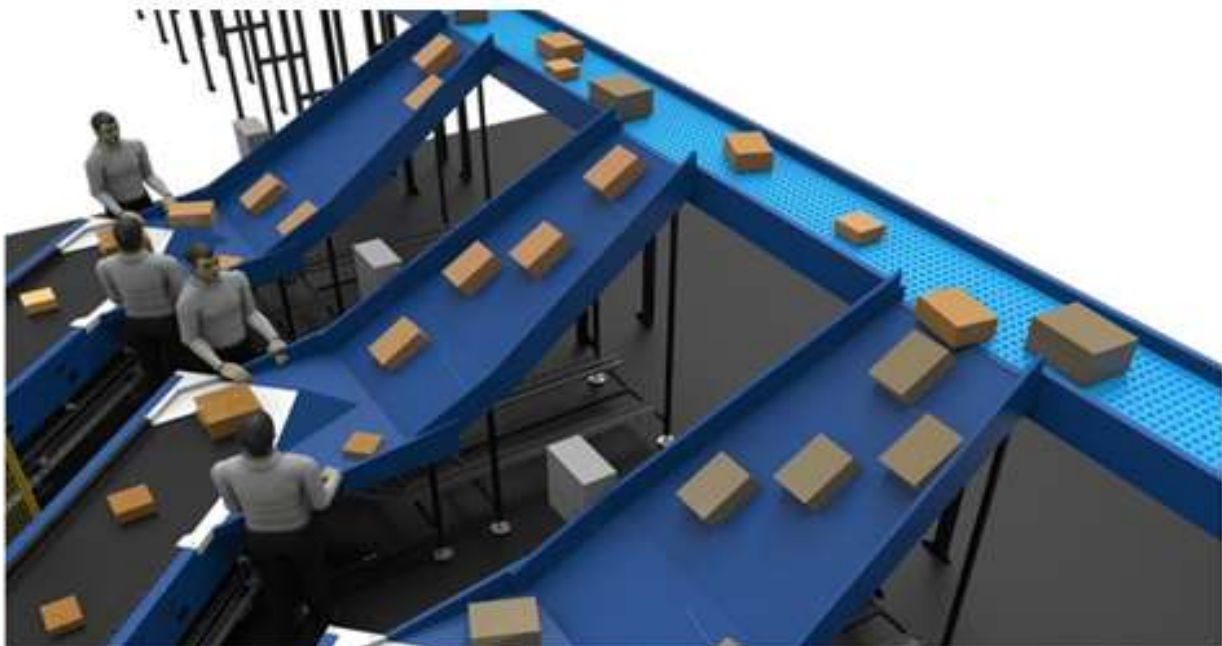
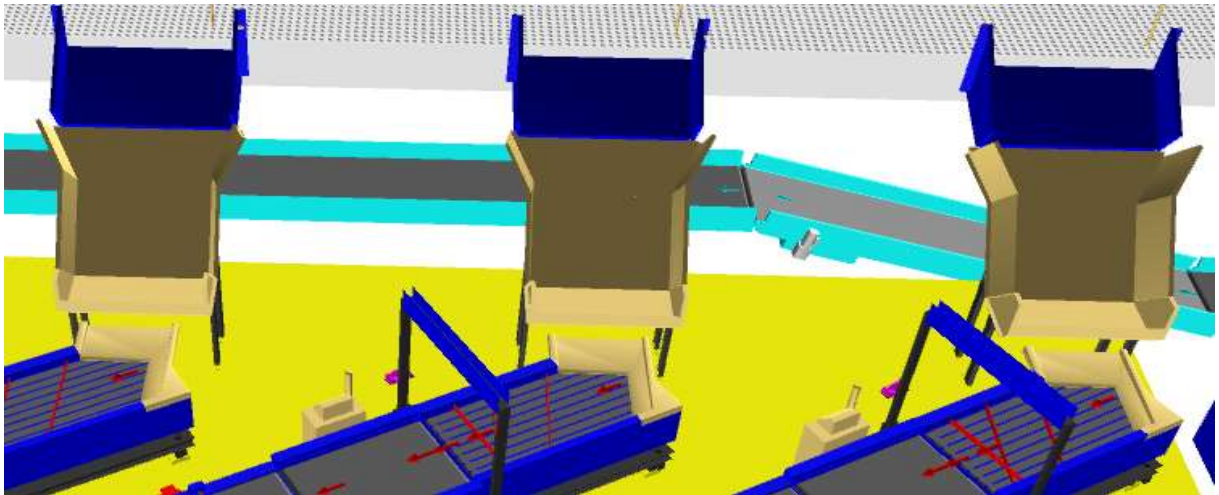
The ARB based Volume Distribution system segregates the incoming bulk flow of shipments from the infeed conveyor. It discharges the parcels onto the distribution sliding chutes on pre-defined logic. The left-over shipments are re-circulated in a Loop.
Belt Make – SystemPlast



j. VDS Chutes

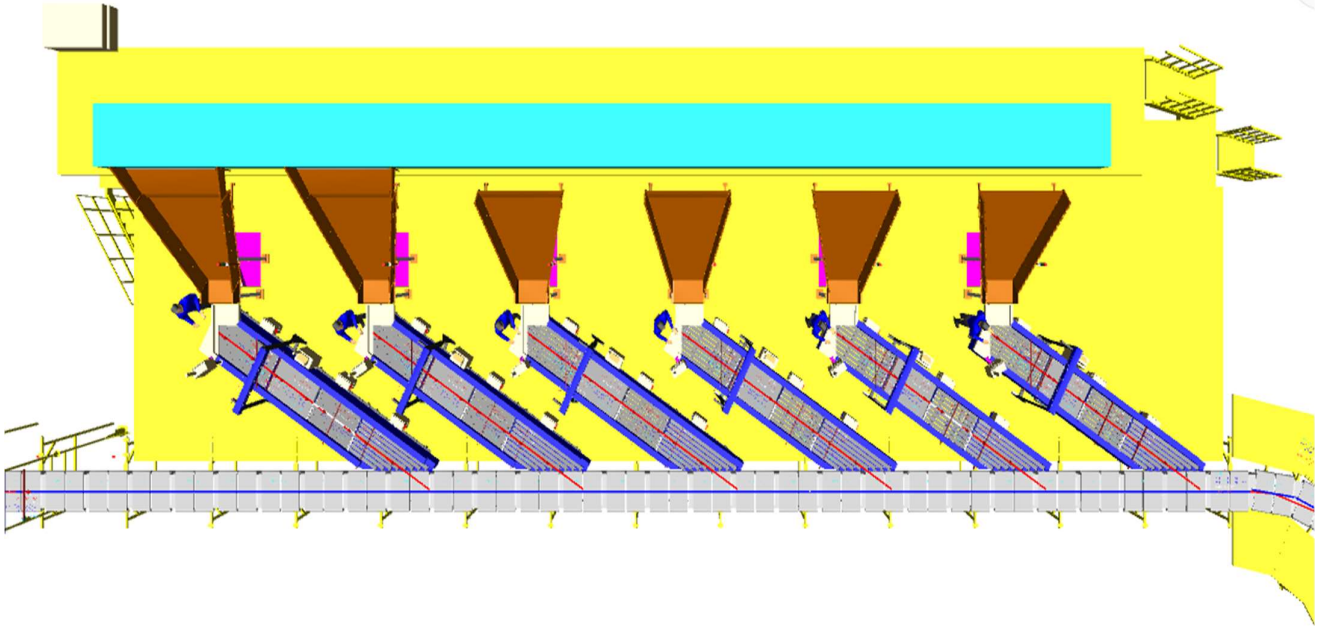
These are erected after the VDS to accumulate incoming parcels to be segregated, hence leads to the effective buffer for the Induct operators. These chutes are equipped with Sensors that give feedback to VDS.

Chute Full Sensors are also provided to alert the system that the chute is full.



k. Feedlines

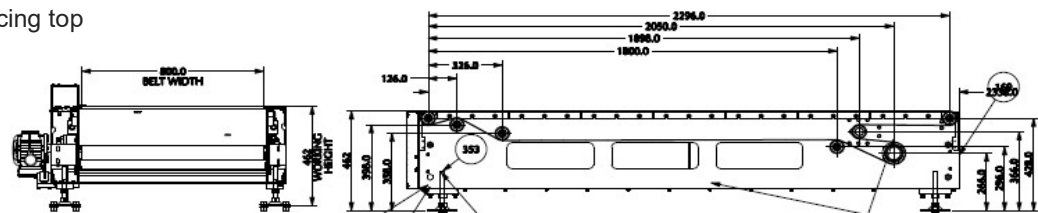
The System has total 12 Automatic Feedlines with Designed IPP of 2000 PPH



1. Each Feedline has Four Conveyor Modules- 1 Orientation Conveyor, 1 Buffer conveyor, 1 Weighing and 1 angle merge conveyor.
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Parcel on the conveyor and prepare the Cross-Belt Cell for receiving the parcel with high precision.
3. Over the feedline, parcels are evenly spaced and smoothly transferred to the main Loop
4. Size Gauge will be provided at Induct Stations to help Operators to determine the Oversized and Undersized Parcels.

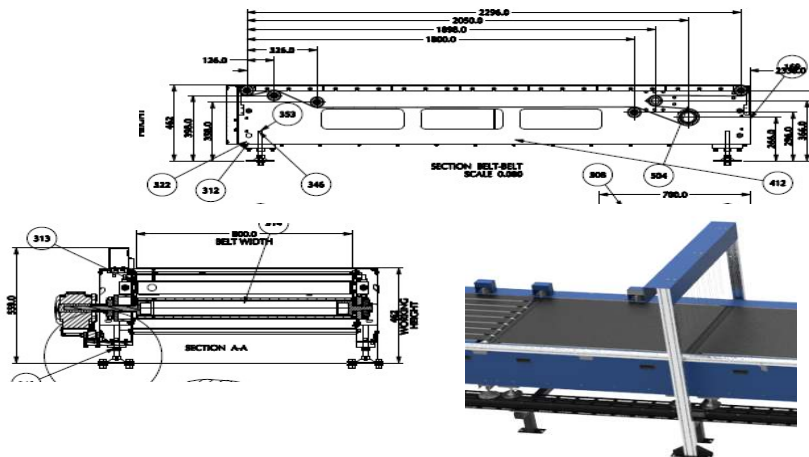
i.Orientation Conveyor

Operators slide the shipments from VDS chutes onto this conveyor and release them ensuring barcode facing top



ii.Buffer Conveyor

High-speed conveyor, used to create appropriate space between parcels/boxes and synchronize the infeed into the Sorter



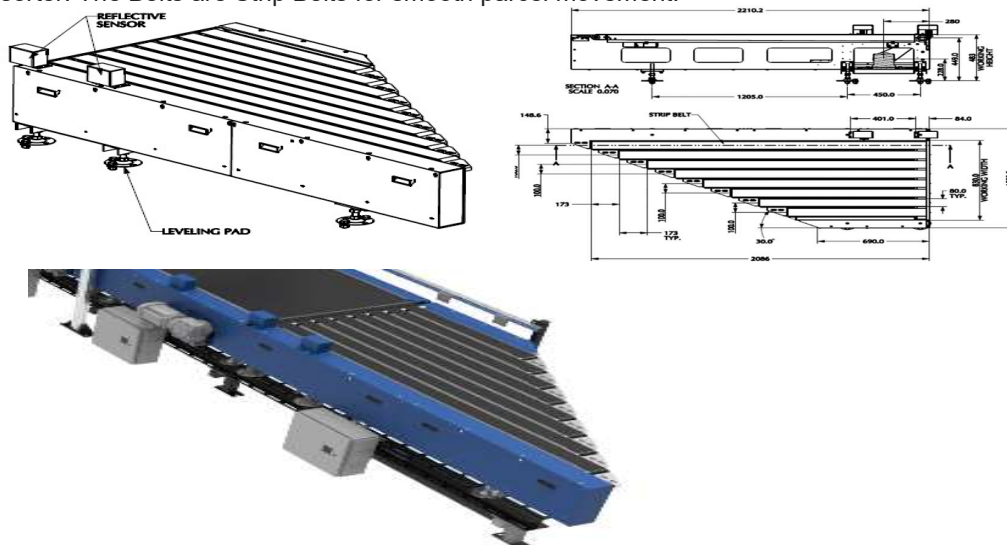
iii. Weighing Conveyor

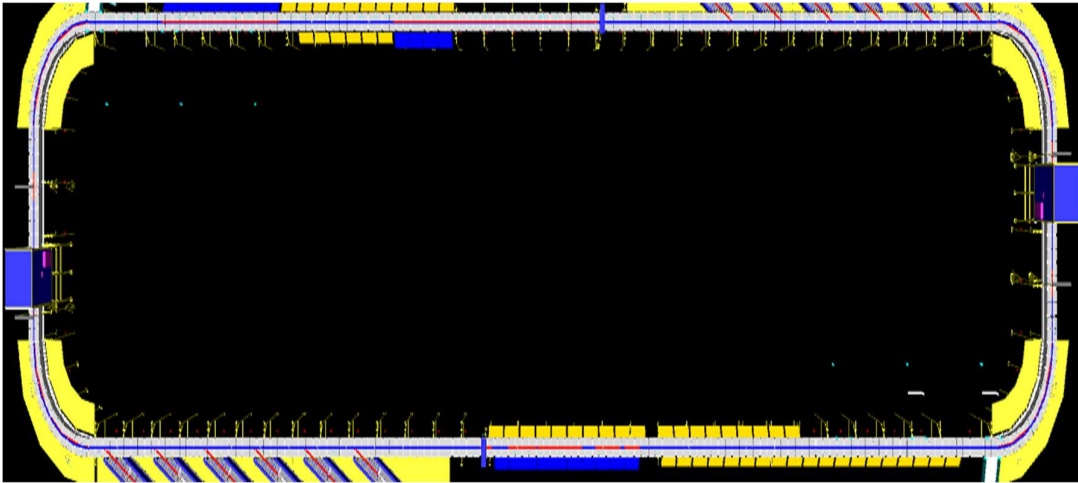
High Accuracy weighing scale for inline weight scanning.



iv. Intelligent Merge

This is 30° triangular Dual Belt high-speed conveyor used for inducing parcel/boxes directly on to the sorter. The Belts are Strip Belts for smooth parcel movement.



v.Main Loop

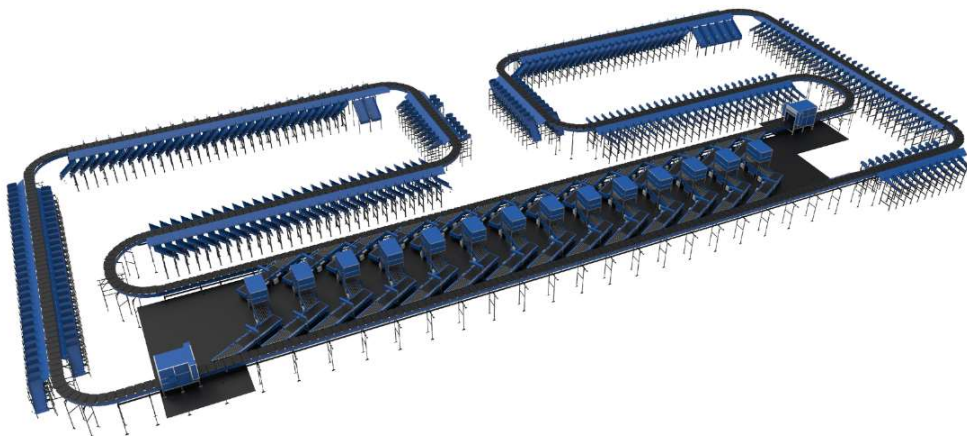
Proposed CBS loop is running at 3400mm height from base level. System is provided with dual belt 1175 mm pitch and Belt Size of 1175 x 800mm.

As Parcels pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches respective chute, carrier carrying the shipment for the chute, will actuate and drop the shipment in the assigned chute.

4.3 Cross Belt Sorter -

Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



4.4 Servo Roller

High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.



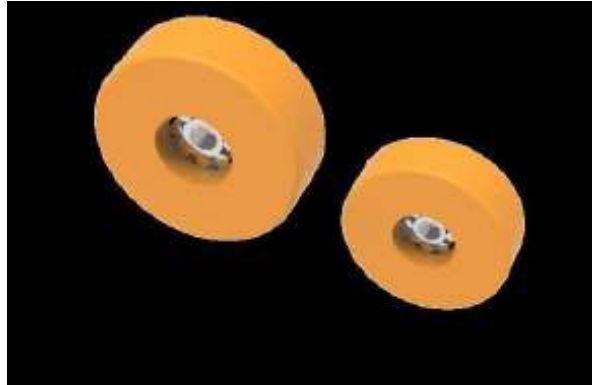
4.5 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.



4.6 Carrier Wheels

- Carrier wheels that are thoroughly tested and proved for long life cycle



4.7 Non-contact based linear motor drive

Extremely **High Energy Efficient** Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed



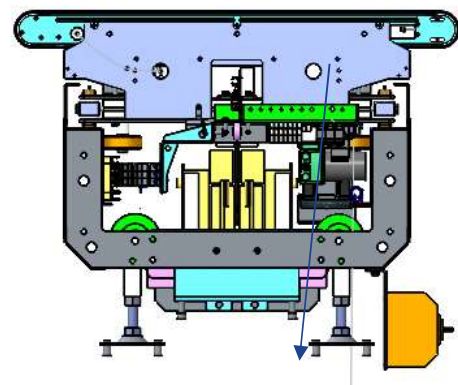
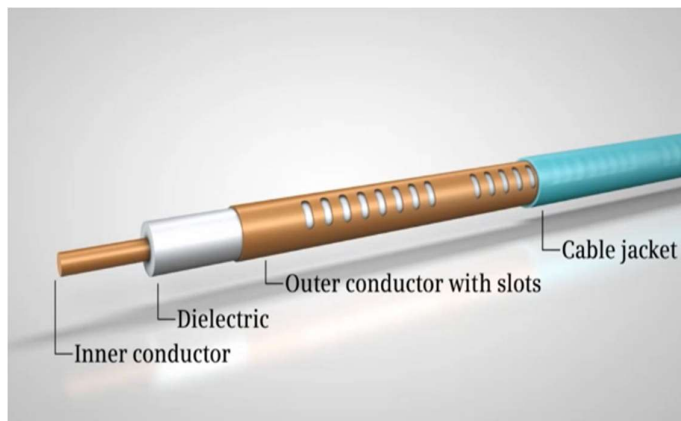
4.8 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.



4.9 Data Transmission

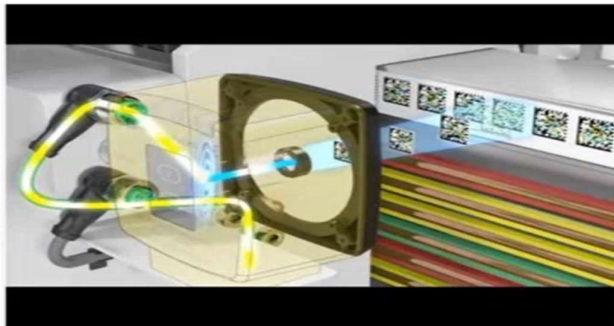
The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.



Leaky wave cable position

4.10 Carriers positioning System

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



4.11 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	Dual belt Carrier Loop CBS
Sorter Speed in m/s	Up to 2.0 m/s
Sorter Loop Length	~200.4 m
Length of the Sorter	Layout Attached
Width of the Sorter including Chutes	Layout Attached
Sorter Actuation Technology	Electric
Chutes	
Chutes + Rejection	102 + 2
Height of Sorter-Lower Deck -	~3400 mm
Track Material & Type of Design to move the Carrier trolleys	~Steel Track
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	1175
No. of Carriers	171 Nos.
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during DAP
Induction Mechanism to the Sorter	
Feedlines (Automatic)	12 Nos
No. of Barcode Scanners (ICR type)	02 Nos
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	02 Nos
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

4.12 Scanning & Sensing on Main Sorter Loop

1. Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan parcel 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time.

Handheld Barcode Scanners are provided to run the machine on Semi- Automatic Mode, if required

VMS 5200 quoted as option for Dimensioning of Parcels.



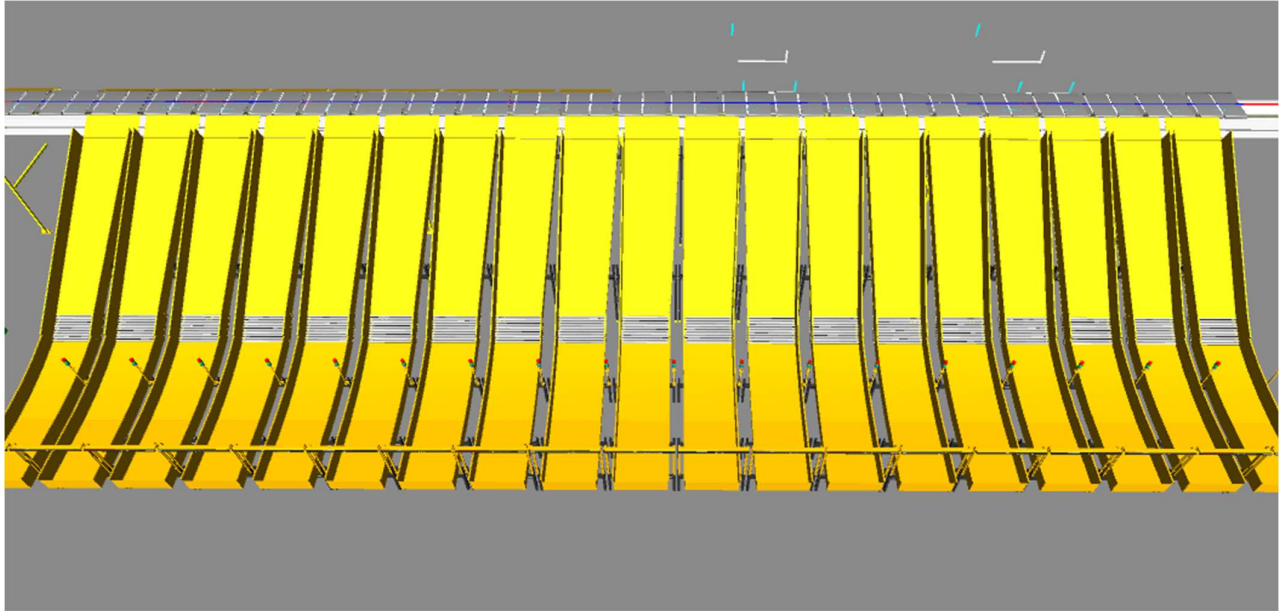
2. Parcel Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple parcel positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Parcel on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).

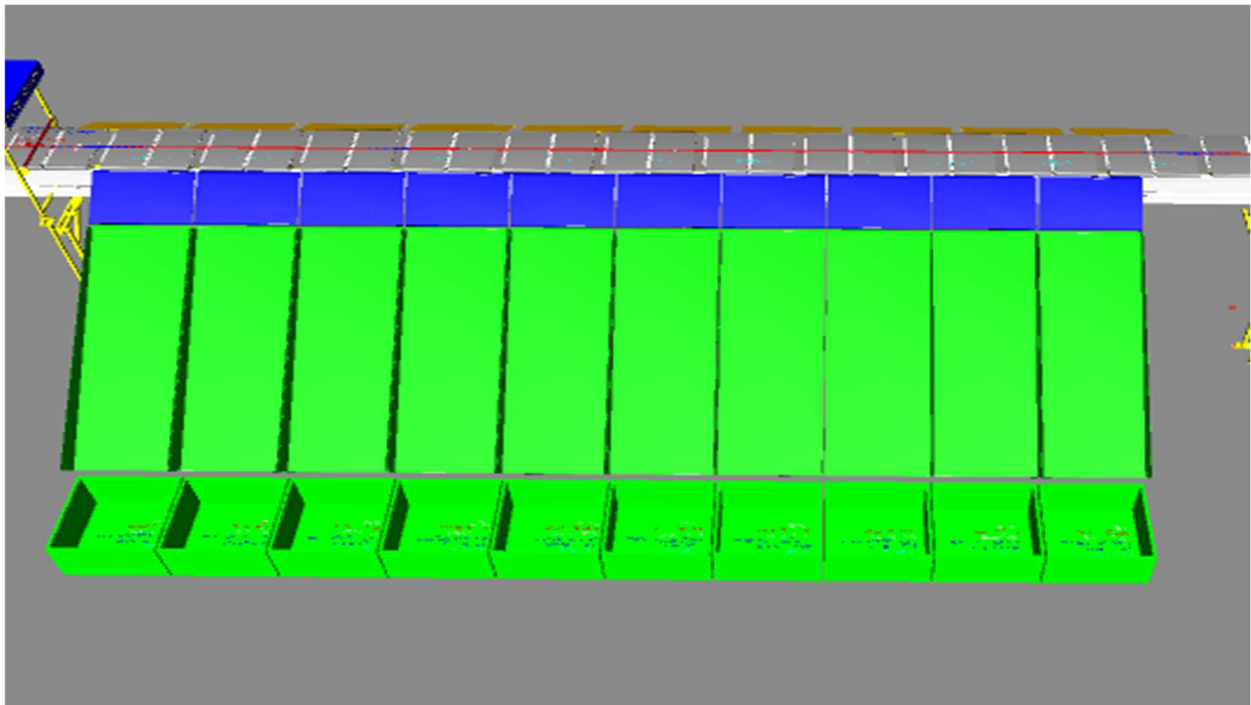


I. Output Chutes

i.High-Capacity Chutes



ii.Direct Drop Chutes



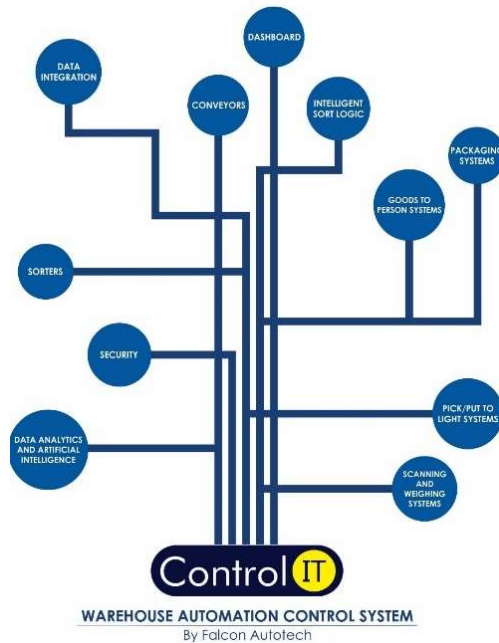
11. Proposed System Technical Details/BOM

Mechanics		
Qty.	Description	Value
1	Conveyor Automation 34 x Transport Conveyor; L-1332m PVC belt conveyor; L-876m Modular belt conveyor & Turn; L-305m Accumulation conveyor; L-366 25 x Merger 16 x Gapper 03 x sorter for inbound 07 x sorter for singles 08 x sorter for multis 12 x ARB conveyors & Divert point; L-50mtr 04 x Aligning conveyor; L-4m.	Width- 600/800/1000/1200 mm
Controls		
1	Controls, IT and Wiring Software Package Control Panels Sorter Control Electrical & Control Hardware Switches Drive panels Operator Panels Internal & Field Wiring	Included

12. High Availability Server Specifications

S. No	Configuration	Role	Qty	Purpose
1	Dell PowerEdge T440 (Chassis with up to 8, 3.5" Hot Plug Hard Drives) Tower Server Trusted Platform Module 1.2 Intel Xeon Silver 4110 2.1Ghz, 8C/16T, 9.6GT/s 2UPI, 11M Cache, Turbo, HT (85W) DDR4-2400 40 GB (2 x 16GB, 8 GB) RDIMM 2666MT/s Dual Rank 2 x 1TB SATA 7.2K 3.5" 6Gbps Fully assembled (Hot Plug) Hard Drive PERC H330 Integrated RAID Controller DVD +/-RW, SATA, Internal Single Hot-Plug Redundant Power Supplies (1+0) 495W Dual-Port 1GbE On-Board LOM iDRAC9 Express 3Yrs Basic Warranty - NBD	Node 1 - Prod Server	1	Failover Cluster is configured on this system. If this node fails, then the VM shifts from this node to another and vice versa
2	Dell PowerEdge T440 (Chassis with up to 8, 3.5" Hot Plug Hard Drives) Tower Server Trusted Platform Module 1.2 Intel Xeon Silver 4110 2.1Ghz, 8C/16T, 9.6GT/s 2UPI, 11M Cache, Turbo, HT (85W) DDR4-2400 40 GB (2 x 16GB, 8 GB) RDIMM 2666MT/s Dual Rank 2 x 1TB SATA 7.2K 3.5" 6Gbps Fully assembled (Hot Plug) Hard Drive PERC H330 Integrated RAID Controller DVD +/-RW, SATA, Internal Single Hot-Plug Redundant Power Supplies (1+0) 495W Dual-Port 1GbE On-Board LOM iDRAC9 Express 3Yrs Basic Warranty - NBD	Node 2 - Prod Server	1	Failover Cluster is configured on this system. If this node fails, then the VM shifts from this node to another and vice versa
3	Intel® Xeon® processor E3-1200 v5 product family, 16 GB RAM, 1 TB * 2HDD, RAID 1, iDRAC8 Enterprise (upgrade), 3 years Onsite warranty.	Domain Server	1	Hyper V will work with Domain System So that we required Domain Server Which will manage the nodes
4	Synology storage with 4 TB HDD	Network Storage Box 1	1	For Creation of Clustered Share Volume, VM will be created on this BOX
5	Synology storage with 4 TB HDD	Network Storage Box 2	1	Backup & Replication of CSV
6	Power Backup	Power Backup	3	UPS for the Servers & Network Boxes
7	Monitor - 19.5 inch	Monitor	1	Monitor
8	KVM Switch	Switch	1	To switch over the display between servers
9	Windows Server 2016 Standard R2 or Higher	Windows License	3	OS

13. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions**Supports various peripheral Equipment Inputs and Outputs**

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

14. Key Component Makes

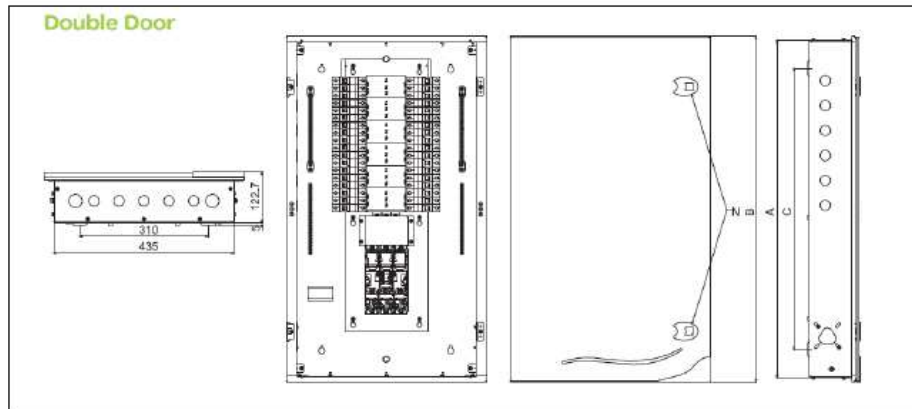
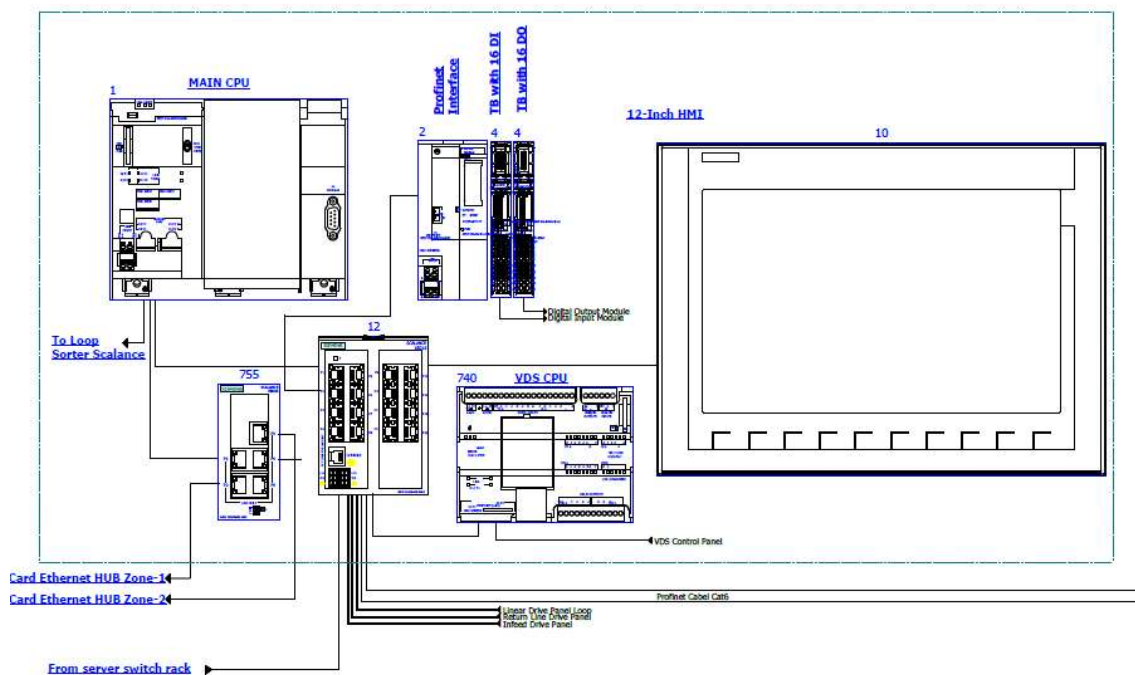
Component	Make
PLC & RIO	Siemens/Omron
VFD	Siemens/Omron/Lenze/AB
Ethernet switch	Siemens/Phoenix Contact
SMPS	Omron/Selec/Meanwell
Line chokes	EMIS/GURU NANAK ELE.
Slim relays	Phoenix contact
Terminals	Wago
Tower lamp + Hooter	Leuze
Utility switch & socket	Anchor
Fan/Air conditioner	Rexnord
Switchgear	Schneider
Main load break switch (CP)	Schneider
Geared Motors	SEW
MDR	Pulse
Cameras	Cognex/Sick

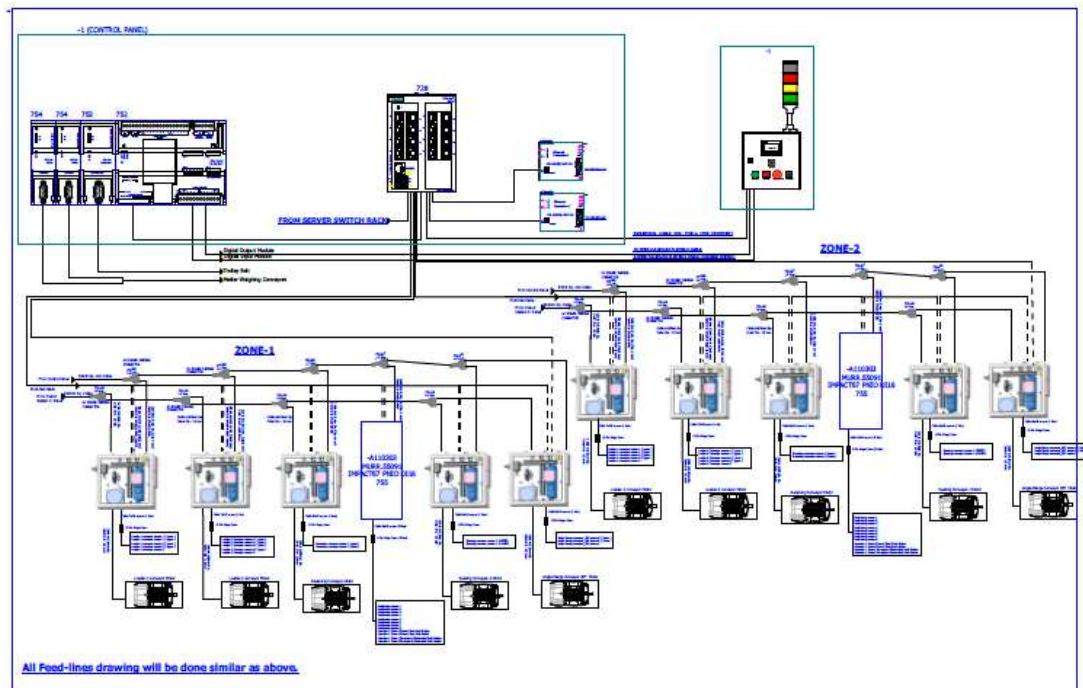
5. Electrical System

Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.

PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

a. Reference Picture of Power Distribution Panel**b. Main Control Panel (Reference)**

c. Induct Stations Control Panel (Reference)**Engines**

Three-phase alternating current motors (Induction) will be used through a frequency converter.

The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.

All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

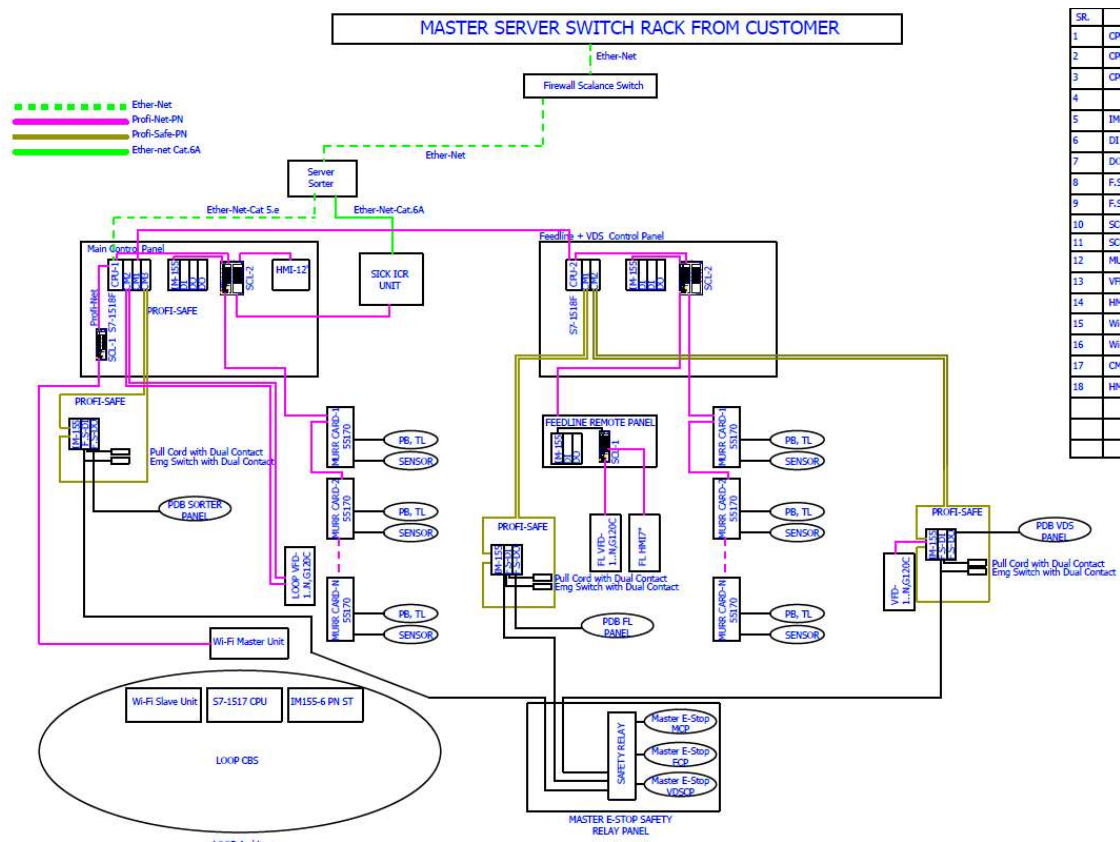
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

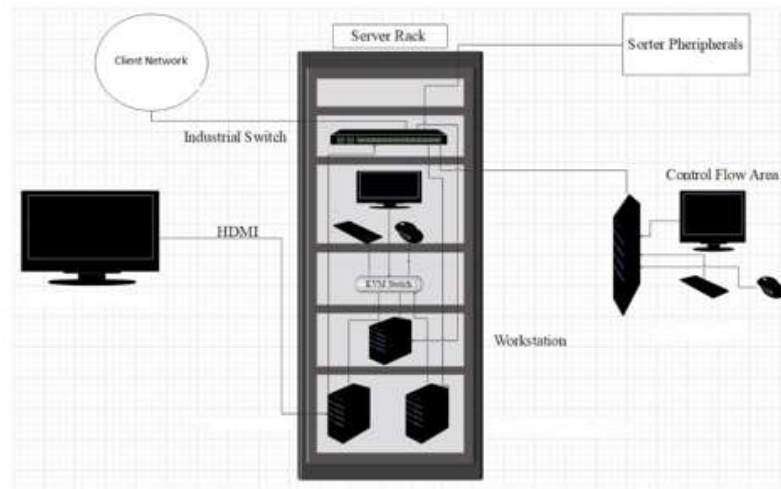
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

6. Controls Architecture



7. IT Architecture (Ref.)

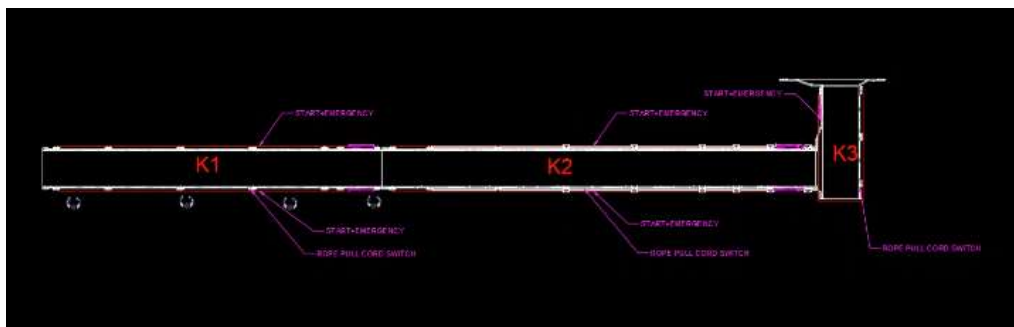


8. Principle of Safety

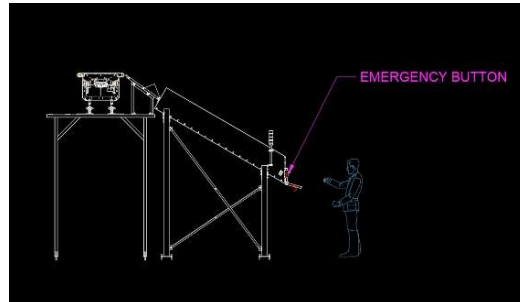
1. E-Stops



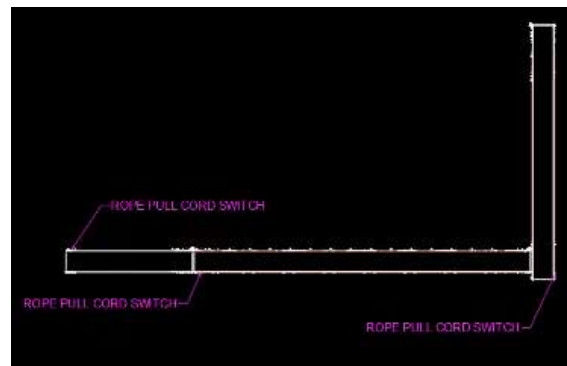
- a. At Every Conveyor Module – Both sides



b. VDS Chutes



2. Pull Cords Switch- PVC Conveyors

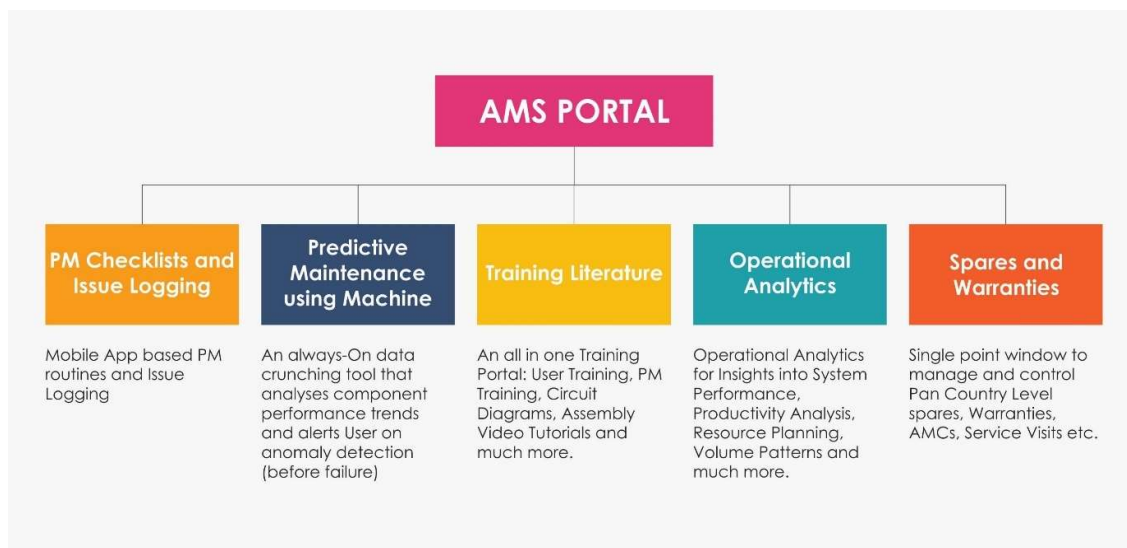


3. Fencing

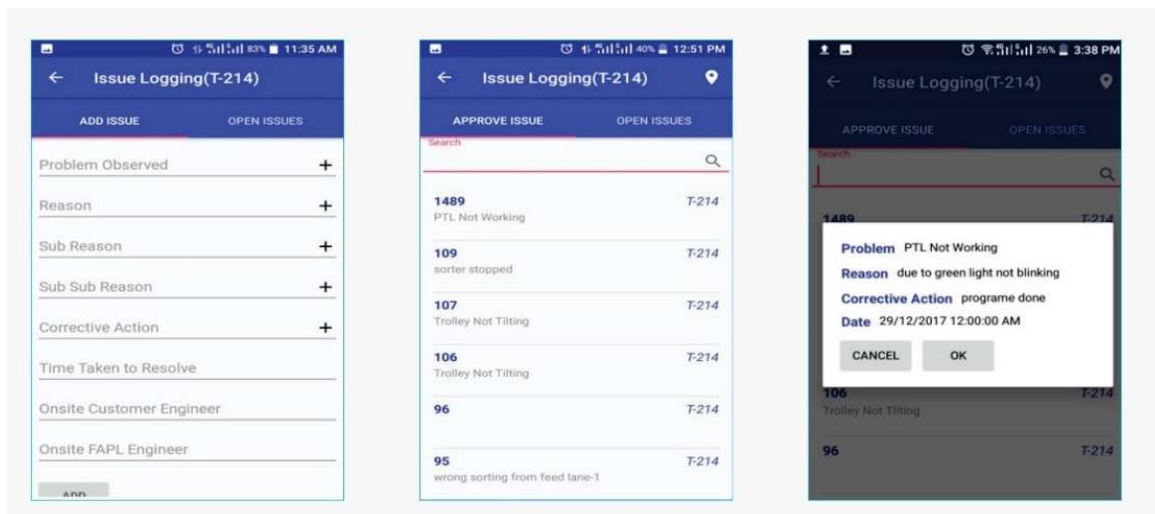
- - a. Between Inducts
 - b. Between Inducts and Sorter



15. Automation Management System (AMS)



a. Issue Logging



b. Preventive Maintenance Check List

CL(T-214)
Selected Date: 2018-02-27

HOURLY(4) DAILY(111) WEEKLY MONTHLY

Operations OK
Collection Bin

Chutes overload

Induction NOT OK
Wide Conveyor

Loading table overload

Rejection NOT OK
Last Collection Bin

Rejection bin overload

PLC Logs OK
PLC

Tray not tilting at last station.

CL(T-214)
HOURLY DAILY(79) WEEKLY MONTHLY

Main Sorter NEW
Tray
Tray Loose

Item Main Sorter
Sub Item master & slave motor
Observation Check for oil leakage mark in gear box
Remarks

NOT OK OK

Main Sorter NEW
slave end
Abnormal Sound

Main Sorter NEW
Trolley Sensors
clean

c. Predictive Maintenance

Notifications(T-214)

Sorter 27/02/2018 12:03:14 PM
Packet between 2 trays after Secondary
Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 12:02:13 PM
Packet between 2 trays after Secondary
Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 11:54:08 AM
Packet between 2 trays after Secondary
Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 11:52:06 AM
Packet between 2 trays after Secondary
Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 11:52:05 AM
Packet between 2 trays after Secondary
Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 11:44:02 AM
Packet between 2 trays after Secondary
Repeated 10 Times, From 120 Seconds.

Sorter 27/02/2018 11:42:01 AM
Packet between 2 trays after Secondary
Repeated 10 Times, From 120 Seconds.

Notifications(T-214)

Trolley[7932888704|0033] 2018 11:49:19 AM
Extra Loop Taken. AWB= 7932888704 and
Extra Loop 0033 Repeated 5 Times.

Feedline[2] 21/03/2018 11:41:52 AM
Same NOBarcode Repeated 10 Times.

Feedline[1] 21/03/2018 11:31:39 AM
Same NOBarcode Repeated 10 Times.

Feedline[2] 21/03/2018 11:24:34 AM
Same NOBarcode Repeated 10 Times.

Feedline[1] 21/03/2018 11:14:34 AM
Same NOBarcode Repeated 10 Times.

Station[16] 21/03/2018 9:35:55 AM
Chute Detection

Station[16] 21/03/2018 9:35:54 AM
Chute Detection

Station[7] 21/03/2018 9:35:53 AM
Chute Detection

d. Training Portal

Employee Training Admin

Training Type: Search

Show All Scheduled Completed

Employee Name	Training Type	Subject	Notes	Training Taken To	Actual Start Date	Actual End Date	Rating
Shank Rupa G.171	Product Training	TB Tray Sorter/TB Tray Loop Sorter	Sorter Sorter	Unit Chockery E.			2
Shank Rupa G.172	Product Training	System PTL Conveyor Belt Conveyor		Unit Chockery E.			2
Shank Rupa G.173	Product Training	BEST CONVEYOR WITH REJECTION AND SORT		Unit Chockery E.			2
Shank Rupa G.174	Product Training	CONVEYOR SORTER SYSTEM		Unit Chockery E.			2
Shank Rupa G.175	Product Training	TB Tray Sorter SHEEP SORTER		Unit Chockery E.			2
Shank Rupa G.176	Product Training	CONVEYOR SORTER SYSTEM		Unit Chockery E.			2
Shank Rupa G.177	Product Training	TB Tray Sorter SHEEP SORTER		Unit Chockery E.			2
Shank Rupa G.178	Product Training	CONVEYOR SORTER SYSTEM		Unit Chockery E.			2
Shank Rupa G.179	Product Training	TB Tray Sorter SHEEP SORTER		Unit Chockery E.			2
Shank Rupa G.180	Product Training	CONVEYOR SORTER SYSTEM		Unit Chockery E.			2

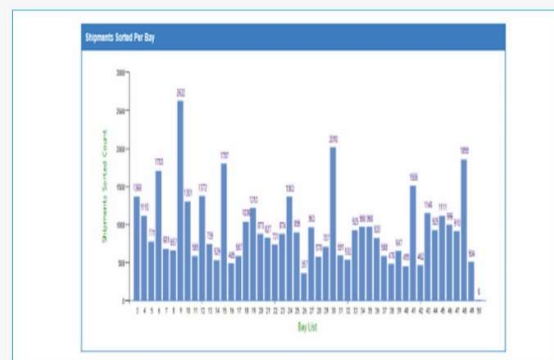
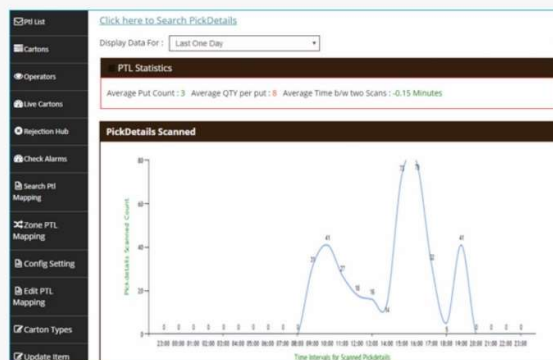
Training Material Admin

Training Type: Document Name: Video Name: Search

Select Page Size

Training Type	Subject	Document Name	Training Document	Video Name	Training Video	Time Limit	Rating
Comprehensive Training	KIR TRAINING	SPACE MANUAL	download		2	2	2
Comprehensive Training	SPD TRAINING	POWERLEX AIR MANUAL	download		2	2	2
Comprehensive Training	PTL	PTL INSTRUCTION MANUAL	download		2	2	2
Comprehensive Training	PTL	operational manual	download		2	2	2
Product Training	TB Tray Loop Sorter	1-214 Sorter Sorting manual	download		2	2	2
Maintenance & Troubleshooting	TB Tray Sorter	T-214 MAINTENANCE MANUAL	download		2	2	2
Product Training	TB Tray Loop Sorter	1-214 SORTER MANUAL	download		2	2	2
Product Training	TB Tray Loop Sorter	1-214 REDUCTION LINE MANUAL	download		2	2	2
Department Procedures	Field & Panel Wiring	Electrical 01	download		20 mins	2	2

e. Operational Analysis



16. Program Organisation

Lead time for entire project would be 36 to 40 weeks Ex-works Greater Noida, India from the date of PO and advance receipt.

a. Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Amazon
3. Scheduling of the resource management.
4. Management of the risks and opportunities.
5. Management of the list of anomalies or reservations.

17. Commercial Terms

The Total Price is to be understood

- Ex Works – Greater Noida, India
- Freight charges shall be extra at actual
- GST will be charged extra @ 18%
- Including Packing and Forwarding
- Duty unpaid
- Installation and Commissioning: Mentioned above.
- Price is valid for 60 days from the date of proposal

Our Price does not include taxes, and duties, import duties, charges, fees, taxes etc. charged by authorities outside India for deliveries and services. Same shall be borne & paid by client itself.

*Material Unloading & Laydown area is in customer scope

a. Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 20% Advance along with PO

Stage 2: 60% + Taxes Against dispatch of the material

Stage 3: 10% against Installation

Stage 4: 10% against PBG

LD Clause- 0.5% per week of delay subject to a maximum of 5%

18. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use) and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

19. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms- If not specified
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.